

TrES-1b

R-1616

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_140614 · created 2026-08-02T03:53:31+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456822.8891 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.076 +/- 0.031
Transit depth (Rp/Rs)^2	0.57 +/- 0.46 [%]
Orbital Inclination (inc)	85.02 +/- 1.46
Airmass coefficient 1 (a1)	0.6097 +/- 0.0097
Airmass coefficient 2 (a2)	0.49 +/- 0.23
Scatter in the residuals of the lightcurve fit is	1.58 %
Best Comparison Star	#3 - [484, 170]
Optimal Method	PSF photometry
Transit Duration (day)	0.033 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.076 ± 0.031 vs 0.1358 (deviation 1.31σ)
Duration fit vs published	0.033 d vs 0.1042 d (deviation 2.2σ)
Mid-time O-C	-3.0 min
Depth SNR	1.7
Residual scatter	1.58 %

Light curve

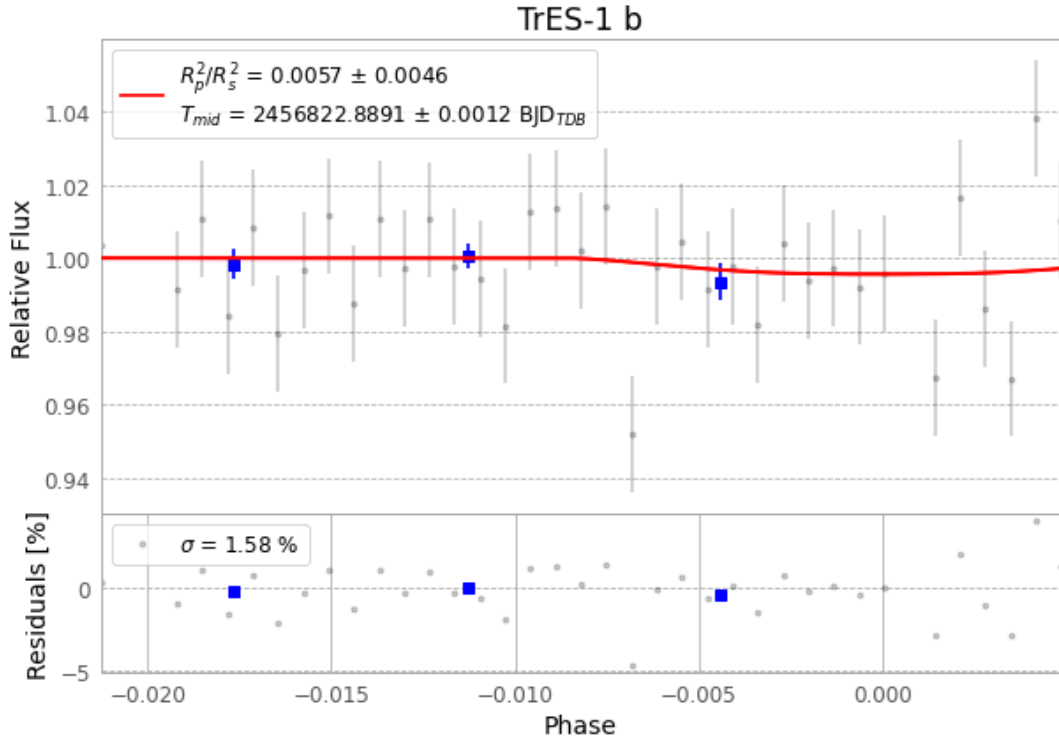


Figure 1 — FinalLightCurve_TrES-1 b_2014-06-14.png (SHA-256 c2b4c307df26f083...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [281, 265], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8df52fb1fd1ce228...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

37 FITS frames (24 MB), TRES-1140614074212.FITS → TRES-1140614093612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-140614.log (892075ec1201...), FinalLightCurve_TrES-1 b_2014-06-14.png (c2b4c307df26...), FinalLightCurve_TrES-1 b_2014-06-14.csv (4227f4fc0789...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 03:53Z.